

Programming, Data Management and Visualization

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July 2025

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1 Data Preparation and Cleaning

1.1 Dataset Overview

The dataset `takehome_health` contains **1,091,328 observations** and **88 variables**. It includes demographic and health-related information, such as age, sex, income, education level, diagnosis history, and health behaviors.

1.2 Checking for Duplicates

Every individual in the dataset should appear only once. To verify this, we used the variable `nhispid`, which uniquely identifies each person in the dataset.

We checked for duplicates using the `duplicated()` function in R, which identifies whether a value has already occurred. By applying it to `nhispid`, we found that **55,643 observations** had duplicate person IDs. This means that these individuals were surveyed more than once, which violates the requirement of one record per person.

To resolve this, we removed all but the first occurrence of each duplicated ID. This ensures that the dataset now includes only one entry per person.

R Code:

```
# Check how many duplicated IDs exist
sum(duplicated(takehome_health$nhispid))

# Remove all duplicated entries, keeping only the first one
takehome_health <- takehome_health[!duplicated(takehome_health$nhispid), ]
```

1.3 Examine and Clean Categorical Variables

Several variables in the dataset are coded as categorical character variables. We examined the consistency of their values using the `table()` function in R.

We focused on five key variables: `sex`, `educ`, `marstcur`, `empstat`, and `region`. Among these, all except `empstat` required cleaning.

The most common issues were inconsistent use of uppercase and lowercase letters, and trailing spaces in category labels. For example, the variable `sex` contained values such as "Male " and " male" which refer to the same category but were treated as different values.

We fixed these inconsistencies using the `trimws()` function to remove spaces and `tolower()` to standardize to lowercase.

R Code Example:

```
# Standardize 'sex' variable
takehome_health$sex <- trimws(takehome_health$sex)
takehome_health$sex <- tolower(takehome_health$sex)
```

This process was repeated for the variables `educ`, `marstcur`, and `region`. The variable `empstat` was already consistently formatted and did not require cleaning.

1.4 Summary of Missing Values

To understand the completeness of our data, we examined missing values for each variable in the dataset. We calculated both the total number of missing values and their percentage relative to the total number of observations.

The table below shows the top 10 variables with the highest percentage of missing values:

Table 1: Top 10 Variables with Most Missing Values

Variable	Missing	Percent (%)
therapyr	1,013,147	97.82
presmed12mo	725,647	70.06
mdsmoktalk	650,764	62.83
mdietalk	595,928	57.54
cholhighcv	579,470	55.95
depfeelevl	539,713	52.11
worfreq	539,602	52.10
worrx	539,570	52.10
depfreq	539,540	52.09
deprx	539,463	52.09

We observe that the highest rates of missing values occur in variables related to health care use and mental health, such as `therapyr`, `presmed12mo`, and `mdsmoktalk`. This suggests that participants were either less likely to report sensitive medical information, or that certain questions were only asked of specific subgroups.

These missing values may need to be addressed (e.g., via imputation or exclusion) depending on the variables included in later analyses.

1.5 Check for Logical Inconsistencies

To ensure the internal validity of our dataset, we tested for logical inconsistencies—cases where variable combinations contradict real-world logic.

First, we examined whether any individuals labeled as **male** were also marked as currently pregnant. Using the variables **sex** and **pregnantnow**, we found **zero such cases**, indicating no inconsistency in that regard.

Next, we checked whether any children under the age of 12 were reported as **Working for pay at job/business**. Again, our check returned **zero cases**, confirming the data was consistent in this context as well.

R Code:

```
# Check for pregnant males
table(takehome_health$pregnantnow, takehome_health$sex)

# Check for working children
sum(takehome_health$age < 12 &
    takehome_health$empstat == "Working for pay at job/business",
    na.rm = TRUE)
```

No action was required, as no logical inconsistencies were identified in these checks.

2 Data Aggregation and Descriptive Statistics.

2.1 Healthcare Access Barriers

To investigate healthcare utilization patterns and access barriers, we analyzed responses related to doctor visits, delayed care due to cost, and affordability of dental, prescription, and mental health care. These outcomes were summarized across subgroups defined by age group, gender, poverty status, income category, and welfare receipt.

For each subgroup, we computed the percentage of individuals who reported having a doctor visit in the past two weeks, delaying care due to financial reasons, and being unable to afford specific types of healthcare.

The table below presents a summary of healthcare access indicators for selected demographic and socioeconomic groups. We report the percentage of individuals experiencing doctor visits, delayed care, and unmet healthcare needs.

Table 2: Sample of Healthcare Access Summary Table

Age Group	Gender	Poverty Status	Income Category	Welfare Receipt	Doctor Visit (%)	Delayed Care (%)
18-44	female	below poverty threshold	\$0-\$34,999	yes	21.53	12.88
45-64	male	at or above poverty threshold	\$75,000+	no	18.40	5.71
...	...	...	...	...	...	...

For transparency and reproducibility, the complete version of this table, including all 249 subgroups and 11 indicators, is provided in the supplementary file `healthcare_summary.csv`.

2.2 Mental Health Access and Utilization

We explored mental health access and utilization by analyzing the percentage of individuals who: (i) saw a mental health professional, (ii) took medication for depression or anxiety, and (iii) reported experiencing symptoms of depression and anxiety. The results were grouped by age, gender, poverty status, income category, and welfare receipt.

The variables used were `sawment`, `deprx`, `depfreq`, and `worfreq`. We recoded frequency responses for depression and anxiety to focus on meaningful symptom reports such as *daily*, *weekly*, *monthly*, or *a few times a year*.

Table 3 shows a preview of the results. The full summary table includes all 249 subgroup combinations and is available as a CSV attachment.

Table 3: Sample of Mental Health Access Summary Table

Age Group	Gender	Poverty Status	Income Category	Welfare Receipt	Saw MH Pro (%)	Took Meds (%)	Depression (%)	Anxiety (%)
18-44	female	below poverty threshold	\$0-\$34,999	yes	4.71	2.13	6.41	5.22
45-64	male	at or above poverty threshold	\$75,000+	no	3.02	1.51	2.98	1.76
...	...	...	...	...	...	...	...	...

Note. The complete table (249 subgroups, 9 columns) is provided in the supplementary

file `mental_health_summary.csv`. It details patterns in mental health service usage and reported symptoms across the population.

2.3 Chronic Disease Indicator

To examine the prevalence of chronic illness across subpopulations, we created a binary indicator variable, `chronic_disease`, which equals 1 if the respondent reported having at least one of the following conditions: diabetes, heart disease, stroke, asthma, arthritis, or cancer. Otherwise, the indicator equals 0.

We then grouped the data by age group, gender, poverty status, income category, and welfare receipt, and calculated the percentage of individuals in each group with a chronic disease.

Table 4: Sample of Chronic Disease Prevalence Summary Table

Age Group	Gender	Poverty Status	Income Category	Welfare Receipt	Chronic Disease Prevalence (%)
18-44	female	below poverty threshold	\$0-\$34,999	yes	14.83
45-64	male	at or above poverty threshold	\$75,000+	no	21.91
...	...	...	...	...	...

To assess whether poverty status is significantly associated with the prevalence of chronic disease, we conducted a Pearson chi-squared test of independence. The test yielded a highly significant result:

$$\chi^2(1, N = n) = 153.3, \quad p < 0.001$$

This indicates a statistically significant association between poverty status and chronic disease, with higher prevalence observed among individuals below the poverty threshold.

Note. The full results, including all subgroup combinations ($n = 249$), are provided in the supplementary file `chronic_summary.csv`.

3 Visualisation

This section presents three figures illustrating healthcare patterns in the population. Each plot is generated using cleaned data and is accompanied by a detailed interpretation. Relevant R code used to produce the plots is included

3.1 Relationship Between Age and Delayed Healthcare

Figure 1 visualizes the relationship between age and the probability of delaying medical care due to cost, separately for individuals below and at/above the poverty threshold.

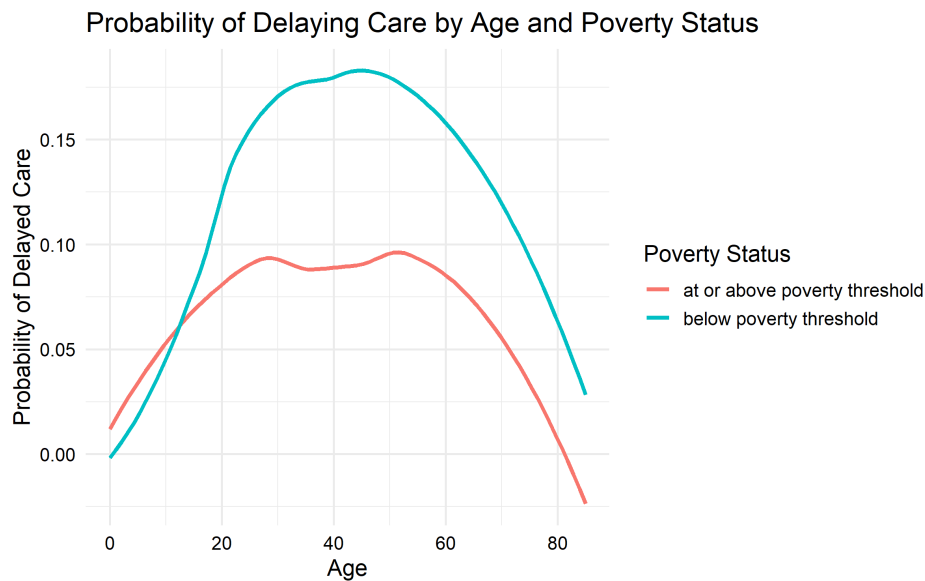


Figure 1: Probability of Delaying Care by Age and Poverty Status

The figure indicates that the likelihood of delaying healthcare rises during early adulthood, peaks around middle age, and declines thereafter. Respondents living below the poverty threshold consistently show a higher probability of delaying care, across nearly all age groups. This suggests that financial barriers to timely medical access disproportionately affect poorer individuals, especially during their working-age years.

3.2 Chronic Disease Prevalence by Age Group and Income Category

Figure 2 shows the prevalence of chronic diseases across different age groups, disaggregated by income category. The data were based on a random sample of the full dataset to ensure performance and clarity in visualization.

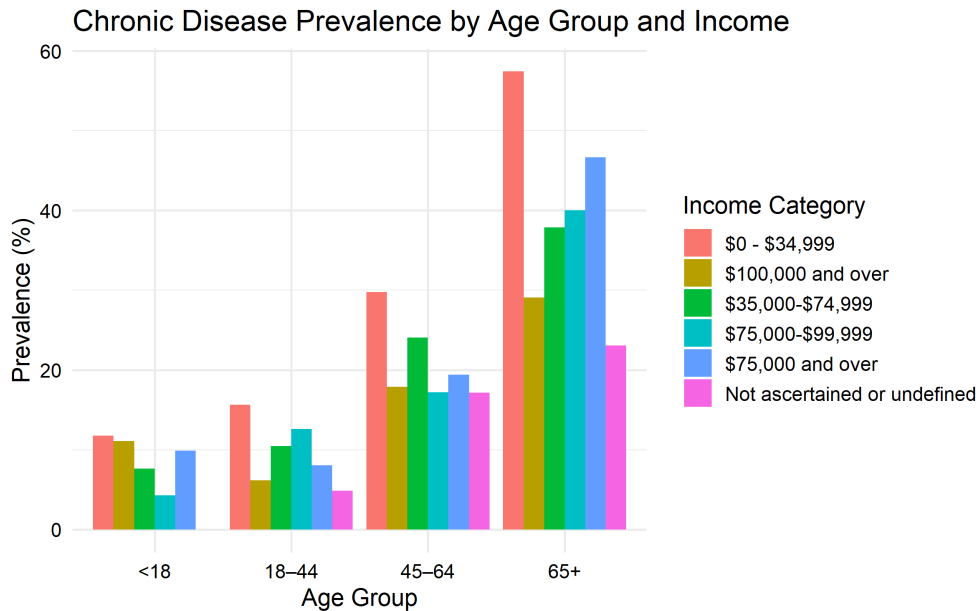


Figure 2: Chronic Disease Prevalence by Age Group and Income Category

The figure illustrates that chronic disease prevalence increases with age, especially after 45 years. Across all age groups, individuals in the lowest income category (\$0-\$34,999) consistently report the highest rates of chronic conditions. In contrast, respondents with higher income levels show lower prevalence. These results suggest a strong association between lower income and chronic disease burden, particularly among older adults.

3.3 MultiPanel Figure – Healthcare Utilization by Demographics

To explore how healthcare utilization varies by key demographic and socioeconomic dimensions, we construct a multi-panel figure showing the percentage of respondents who visited a doctor in the past two weeks. The panels are stratified by education level, while the bars are grouped by employment status and colored by gender.

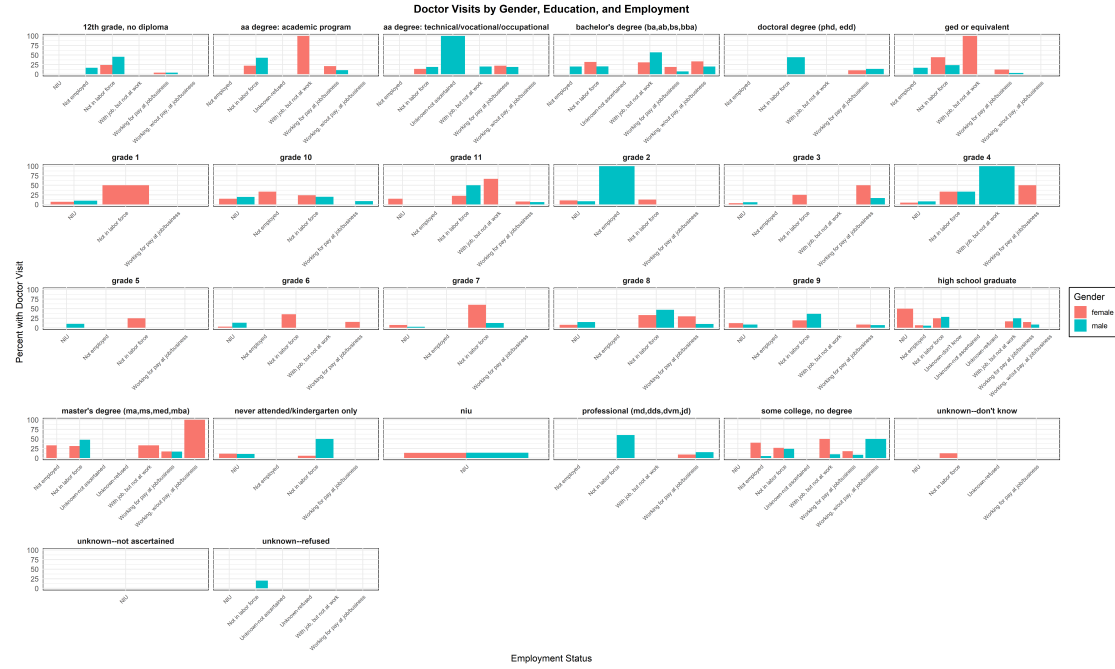


Figure 3: Doctor Visits by Gender, Education, and Employment Status

Figure 3 shows the percentage of respondents who had a doctor visit in the last two weeks, segmented by gender, education level (each facet), and employment status (x-axis). Bars are colored by gender: female in red and male in blue.

We observe substantial variation in healthcare utilization across education and employment groups. In many education levels, females report slightly higher doctor visit rates than males. Employment status also influences access: those working for pay or out of the labor force tend to show higher usage than those unemployed.

However, for some low-frequency combinations (e.g., rare education-employment categories), results may be unstable. These trends suggest that both education and employment are important factors influencing healthcare access, with notable gender differences across categories.